ServiceNow Development with Flow Designer

# Auto Ticket Classification using Flow Designer

Project Documentation

## 1. Introduction

**Project Title:** Auto Ticket Classification using Flow Designer

**Team Members:** List team members and their roles.

| Name | Role / Phase |
|---|---|
| KEERTHANA KAVI | Team Leader – Phase 1: Requirement Analysis & Planning |
| SAANJANA T S | Phase 2: Backend Development & Configuration |
| Srija C | Phase 3: Automation using Flow Designer & Email Notification |
| Thilagavathi K | Phase 4: Testing, Validation & Security |
| Harini J | Phase 5: Deployment & Final Documentation |

## 2. Project Overview

**Purpose:** The school IT helpdesk receives many incident requests daily from students and teachers (Wi-Fi issues, projector failures, password problems, slow computers). IT staff currently review each request manually to identify the issue type, pick a category and subcategory, and inform the caller. This is time-consuming, error-prone and not scalable. This project automates ticket classification at the time of creation using ServiceNow Flow Designer, by analysing keywords in the ticket's short description.

**Goals:**

- Automatically classify incidents when they are created.
- Reduce manual effort for IT agents.
- Improve ticket routing efficiency.
- Implement a no-code, easily maintainable solution.

**Features:**

- Automatic assignment of both Category and Subcategory based on keywords in the short description.
- Dependent choice logic between Category and Subcategory.
- Automated email notification to the caller on ticket creation.
- Custom table (Incident WorkFlow) with an auto-numbered ticket ID (INC prefix) for structured, standardised storage.
- Completely no-code, scalable and easy to maintain; packaged as an update set for deployment.

**Keyword-to-classification rules:**

| Keyword in Short Description | Category | Subcategory |
|---|---|---|
| Wi-Fi / Network | Network | Wi-Fi |
| Projector | Hardware | Projector |
| Forgot password | Access | Forgot Password |
| Slow computer | Performance | Slow Computer |

## 3. Architecture

This solution is built entirely on the ServiceNow platform and has no custom code. The MERN-style layers are mapped to the ServiceNow platform as follows.

### Frontend (User Interface)

The ServiceNow form UI for the Incident WorkFlow table, designed with Form Designer. The student only enters the **Caller** and **Short description**; Category and Subcategory are filled in automatically. Subcategory choices are filtered dynamically by the selected Category.

### Backend (Flow Designer Automation)

A flow named **Auto Classify School IT Tickets** (Application: Global) is built in Flow Designer:

- **Trigger:** Record Created on table Incident WorkFlow, condition Category is empty.
- **Flow Logic:** an If / Else If chain that checks whether Short description contains the predefined keywords.
- **Actions:** an Update Record action in each branch sets Category and Subcategory.
- **Email:** a Send Email action sends the confirmation to the caller's email address.

### Database (Custom Table and Schema)

Table label **Incident WorkFlow**, name **u_incident_workflow**, Auto-number enabled (Prefix INC, starting number 500, 5 digits). Fields:

| Field Label | Type | Reference / Choice |
|---|---|---|
| Number | Auto Number | — |
| Caller | Reference | sys_user |
| Category | Choice | Network, Hardware, Access, Performance |
| Subcategory | Choice String | Wi-Fi, Projector, Forgot Password, Slow Computer |
| Short Description | String Choice | — |
| Description | Reference | — |
| State | Reference | New, In progress, On hold, Resolved, Closed |
| Assigned Group | | sys_user_group |
| Assigned to | | sys_user |

**Dependent choice mapping** (Subcategory depends on Category):

| Subcategory | Dependent value (Category) |
|---|---|
| Wi-Fi | network |
| Projector | hardware |
| Forgot Password | access |
| Slow Computer | performance |

# 4. Setup Instructions

## Prerequisites

- A ServiceNow instance (e.g., Personal Developer Instance) with **admin** access.
- Flow Designer / Workflow Studio available (Process Automation).
- Outbound email configured on the instance (for notifications).
- Update set XML file exported from the source instance (for importing the solution).
- A modern web browser.

## Installation

1 **Create an update set:** All → Local Update Sets → New. Name: Project Update Set, State: In progress, Application: Global. Save, then click Make This My Current Set.

2 **Create the custom table:** All → Tables (System Definition) → New. Label: Incident WorkFlow; uncheck Create module; Save. In the Controls tab enable Auto-number and Save again.

3 **Create the fields:** Open the form → Configure → Form Design and add the fields listed in Section 3. Add choices for Category, Subcategory and State.

4 **Configure the dependency:** Right-click Subcategory → Configure Dictionary → Advanced view → Dependent Field tab. Enable Use dependent field, set Dependent on field = Category, and add the dependent value for each Subcategory choice.

5 **Build the flow:** All → Flow Designer → New → Flow. Name: Auto Classify School IT Tickets, Application: Global. Configure the trigger, the If / Else If conditions, the Update Record actions and the Send Email action.

6 **Activate the flow** using the Activate button at the top right.

7 **(Alternative) Import instead of building:** On the target instance go to Retrieved Update Sets → Import Update Set from XML, upload the exported file, then Preview and Commit.

There are no environment variables to configure. The only environment dependency is the instance's outbound email setup.

# 5. Folder Structure

ServiceNow stores configuration as records rather than files. The solution is organised as the following components inside the update set.

### Client (Form / UI Components)

| Component | Description |
|---|---|
| Incident WorkFlow form | Default view with Number, Caller, Category, Subcategory, Short description, Description, State, Assigned group, Assigned to. |
| Form Design layout | Two-column layout created with Form Designer. |
| Choice lists | Category, Subcategory (dependent) and State choices stored in sys_choice. |

### Server (Tables and Automation)

| Component | Description |
|---|---|
| Update Set: Project Update Set | Container for all changes; exported as XML (sys_remote_update_set_*.xml). |
| Table: u_incident_workflow | Custom table with Auto Number (INC prefix) and access role u_incident_workflow_user. |
| Dictionary entries | Field definitions, including the dependent-field setting on u_subcategory. |
| Flow: Auto Classify School IT Tickets | Trigger, If / Else If logic, Update Record actions and Send Email action. |
| System Logs → Emails | Outbound email records used to verify notifications. |

## 6. Running the Application

ServiceNow runs in the cloud, so no local servers need to be started. After activating the flow:

- **Open the application:** navigate to the Incident WorkFlow table (type u_incident_workflow.list in the All menu filter) and click **New**.
- **Create a ticket:** enter the Caller and a Short description, then Submit / Save.
- **Verify:** reload the record; Category and Subcategory are filled in automatically, and an email is generated for the caller.
- **Monitor the flow:** Flow Designer → Executions to see each run of the flow.

## 7. API Documentation

The solution does not define custom REST endpoints. Because the data lives in a standard ServiceNow table, the platform's built-in **Table API** can be used (the user needs an appropriate role). The Flow Designer trigger fires for records created through the API as well.

| Method | Endpoint | Purpose |
|---|---|---|
| GET | /api/now/table/u_incident_workflow | List tickets (supports sysparm_query, sysparm_limit) |
| GET | /api/now/table/u_incident_workflow/{sys_id} | Retrieve a single ticket |
| POST | /api/now/table/u_incident_workflow | Create a ticket; the flow then sets Category / Subcategory |
| PUT / PATCH | /api/now/table/u_incident_workflow/{sys_id} | Update a ticket |

| Method | Endpoint | Purpose |
|---|---|---|
| DELETE | /api/now/table/u_incident_workflow/{sys_id} | Delete a ticket |

**Example request (POST):**

```
{ "caller": "<sys_id of user>",
"short_description": "Projector not turning on" }
```

**Example response (after the flow has run, abridged):**

```
{ "result": {
 "number": "INC00512",
 "short_description": "Projector not turning on",
"u_category": "hardware",
 "u_subcategory": "projector" } }
```

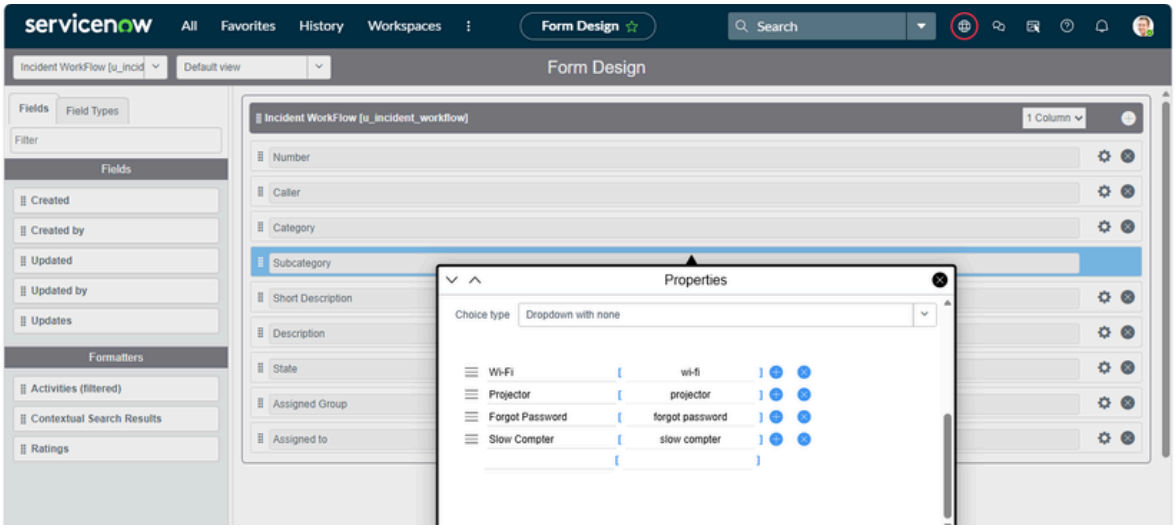
Note: the exact column names depend on how the fields were created (custom fields are prefixed with u_).

## 8. Authentication

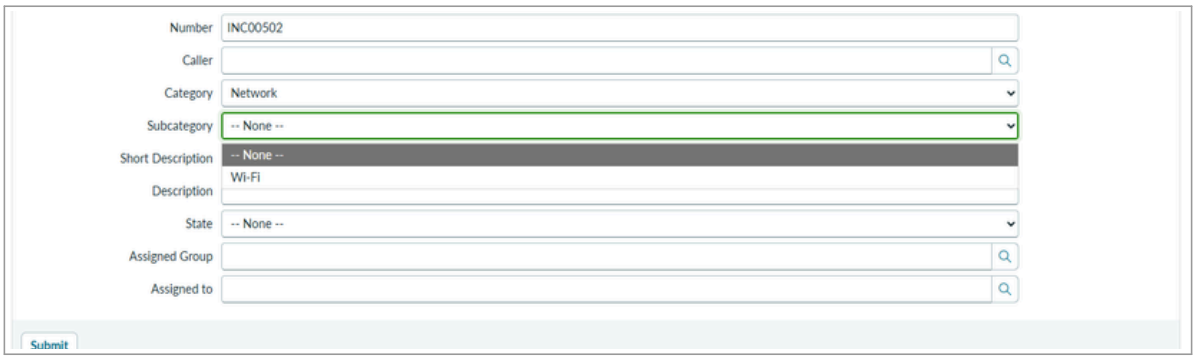
- **Authentication:** users sign in to the ServiceNow instance with their platform credentials (session-based login). API access uses basic authentication or OAuth tokens configured on the instance.
- **Authorization:** role-based access control. When the table was created, Create access controls generated ACLs tied to the role **u_incident_workflow_user**. Only users with this role (and administrators) can read, create or modify tickets.
- **Data integrity:** Caller, Assigned group and Assigned to are reference fields that resolve to the sys_user and sys_user_group tables, so only valid users and groups can be selected.

## 9. User Interface

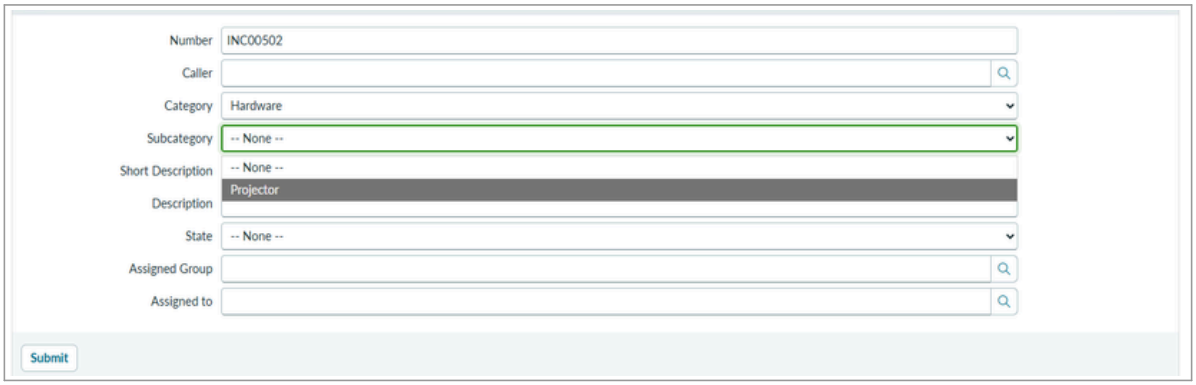
The ticket form shows the Category and Subcategory fields, with Subcategory options filtered by the chosen Category. The screenshots below show the form design and the dependent-choice behaviour.



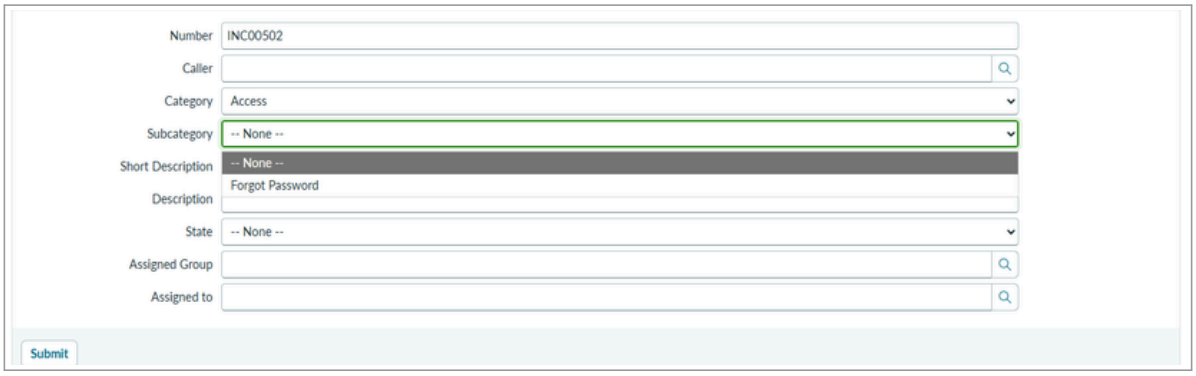

Form Design – Category choices



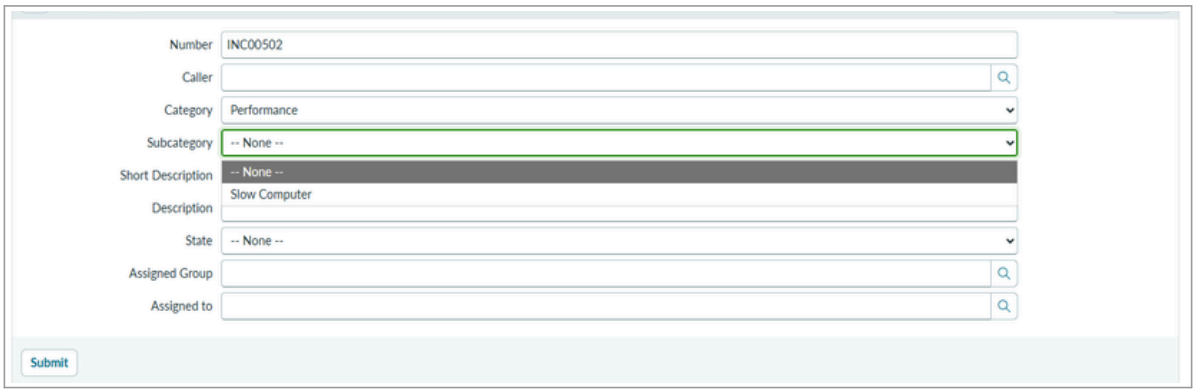

Dependent Subcategory – Category: Network shows Wi-Fi




Category: Hardware shows Projector




Category: Access shows Forgot Password




Category: Performance shows Slow Computer

## 10. Testing

**Strategy:** manual, end-to-end functional testing of the automation, plus data-validation and security checks. **Tools:** the Incident WorkFlow form, Flow Designer, and System Logs → Emails (with Preview Email).

### Data validation checks

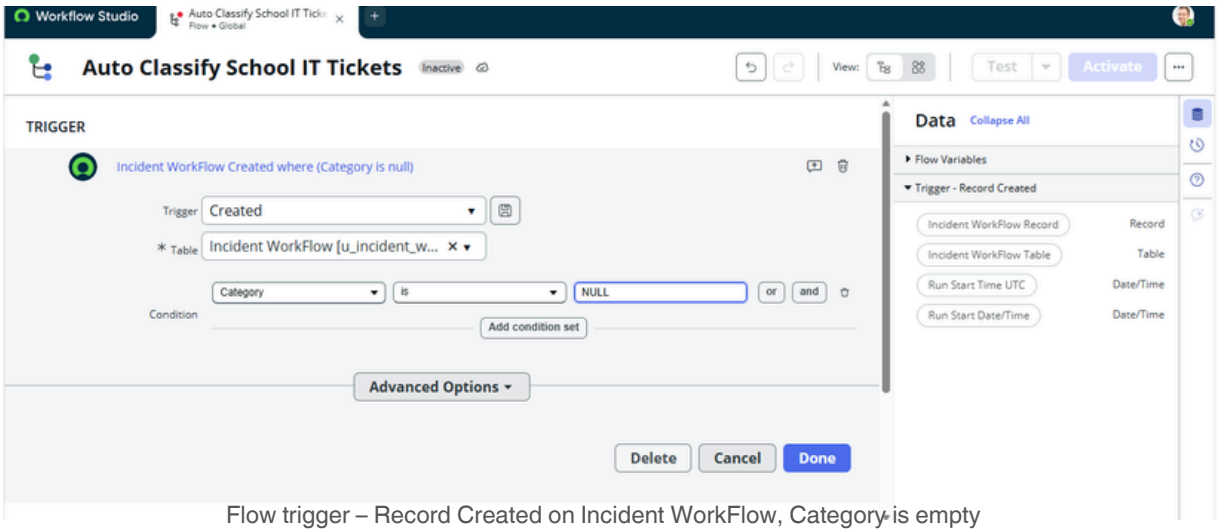
- Mandatory fields are captured correctly.
- Auto-number is generated without duplication.
- Category and Subcategory values are stored accurately.
- Reference fields resolve correctly to the user and group tables.

### Test scenarios

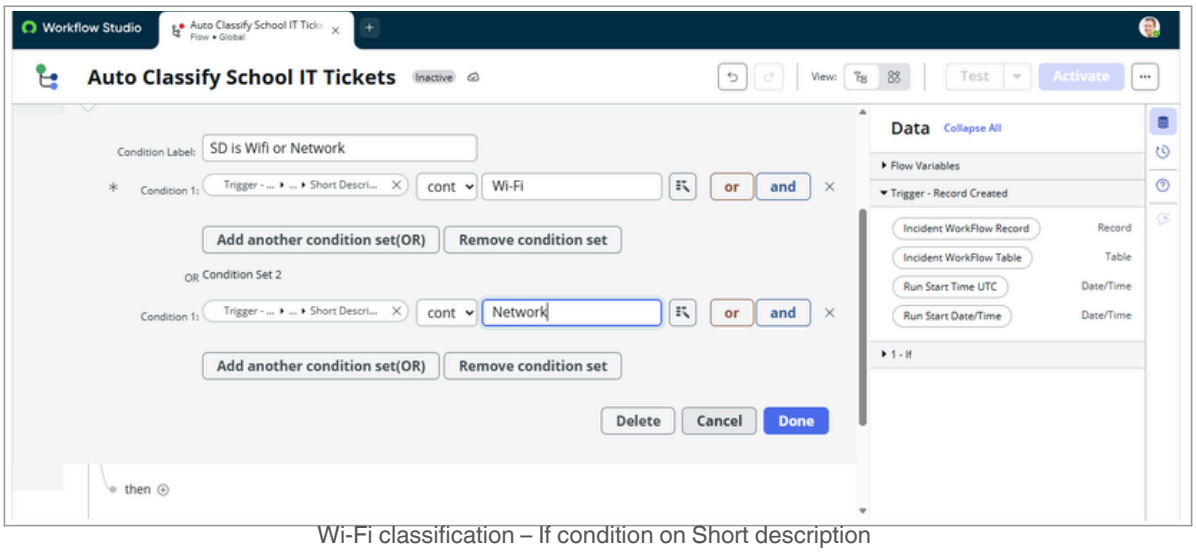
| # | Input (Short description) | Expected Category | Expected Subcategory | Email | Result |
|---|---|---|---|---|---|
| 1 | Wi-Fi / Network issue (e.g., "My college has issue with Network") | Network | Wi-Fi | Sent to caller | Pass |
| 2 | Projector not turning on | Hardware | Projector | Sent to caller | Pass |

**Procedure:** create a new record with a Caller and Short description, save, then reload the form to confirm the fields were filled. To check the email, go to All → Emails (System Logs), search the subject Your Request for the issue has been submitted, open the record and use Preview Email.

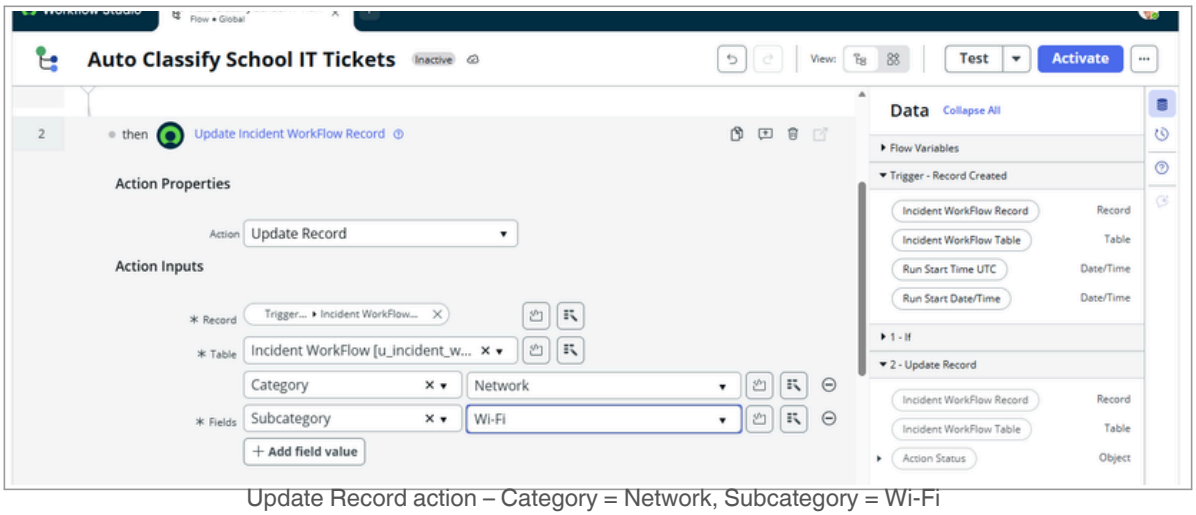
## 11. Screenshots or Demo



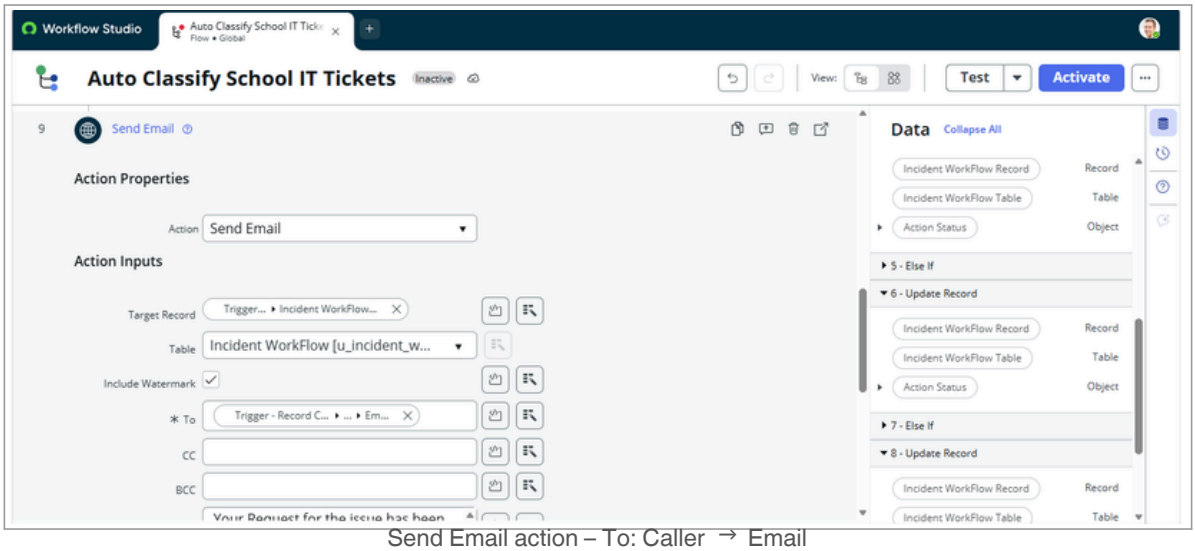

Flow trigger – Record Created on Incident WorkFlow, Category is empty



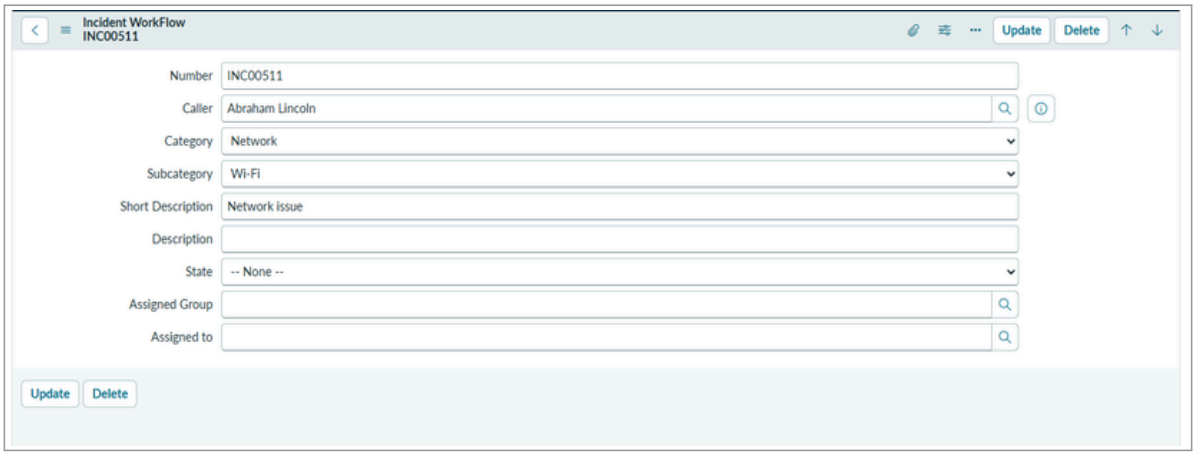

Wi-Fi classification – If condition on Short description



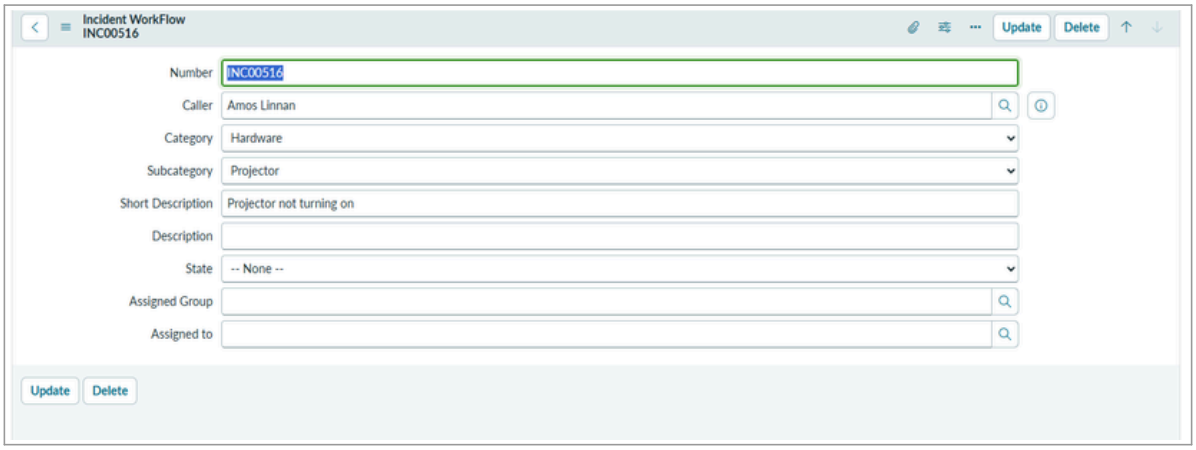

Update Record action – Category = Network, Subcategory = Wi-Fi



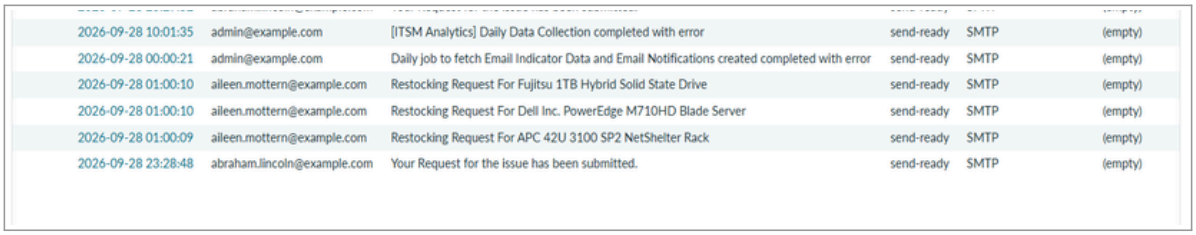

Send Email action – To: Caller → Email




Test scenario 1 – ticket auto-classified as Network / Wi-Fi

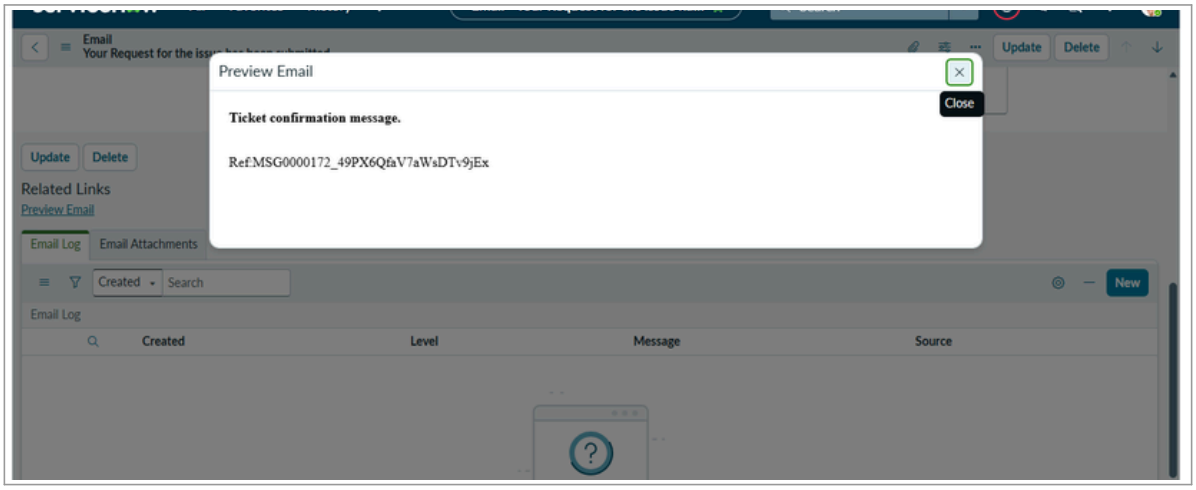



Test scenario 2 – ticket auto-classified as Hardware / Projector



| | | | | | |
|---|---|---|---|---|---|
| 2026-09-28 10:01:35 | admin@example.com | [ITSM Analytics] Daily Data Collection completed with error | send-ready | SMTP | (empty) |
| 2026-09-28 00:00:21 | admin@example.com | Daily job to fetch Email Indicator Data and Email Notifications created completed with error | send-ready | SMTP | (empty) |
| 2026-09-28 01:00:10 | aileen.mottern@example.com | Restocking Request For Fujitsu 1TB Hybrid Solid State Drive | send-ready | SMTP | (empty) |
| 2026-09-28 01:00:10 | aileen.mottern@example.com | Restocking Request For Dell Inc. PowerEdge M710HD Blade Server | send-ready | SMTP | (empty) |
| 2026-09-28 01:00:09 | aileen.mottern@example.com | Restocking Request For APC 42U 3100 SP2 NetShelter Rack | send-ready | SMTP | (empty) |
| 2026-09-28 23:28:48 | abraham.lincoln@example.com | Your Request for the issue has been submitted. | send-ready | SMTP | (empty) |

Email log – confirmation emails to callers




Preview Email – confirmation message

## 12. Known Issues

- **Keyword matching isliteral:** the conditions use contains on exact text (e.g., "Wi-Fi", "Network", "Forgot password", "Slow computer"). Variants such as "WiFi", "wifi" or misspellings may not match, so the ticket stays unclassified.
- **Only the first matching rule applies** (If / Else If chain); a ticket mentioning several issues gets one classification.
- **Keywords listed in the requirement** (e.g., Login, Slow, Hanging) are broader than the conditions currently configured (e.g., "Forgot password", "Slow computer").
- **Email delivery depends on the instance's outbound email configuration;** on a developer instance emails may only appear as send-ready in the Emails log.
- **Subcategory spelling:** keep choice labels consistent (e.g., "Slow Computer") to avoid mismatches in flow values.

## 13. Future Enhancements

- Assignment automation: automatically set Assigned group / Assigned to based on the category.
- SLA tracking and escalation for unresolved tickets.
- Predictive Intelligence to classify tickets using machine learning.
- Expand keyword coverage (synonyms, variants, case handling) and add more categories, ideally using a decision table or a lookup so rules are easier to maintain.
- Add a Service Portal / catalog form for students, and notifications on resolution or state changes.

### Conclusion

The project delivers an end-to-end, no-code automation for the school IT helpdesk. Tickets are classified automatically on creation, dependent choice fields keep data accurate, and callers are notified immediately by email. The design suits small to medium environments such as educational institutions and can be extended with assignment automation, SLA tracking or predictive intelligence.